

THE PLATFORM

One control module. Every power stage.

Power electronics underpins electrification, but above the hobbyist scale, few platforms are open. Sealed firmware and proprietary hardware exclude researchers, educators, and small manufacturers who would otherwise build, study, and improve them. OpenVVF is the open alternative: one reusable control module with swappable power stages for motor drives and bidirectional DC-DC conversion, using IGBTs today with SiC-ready designs.

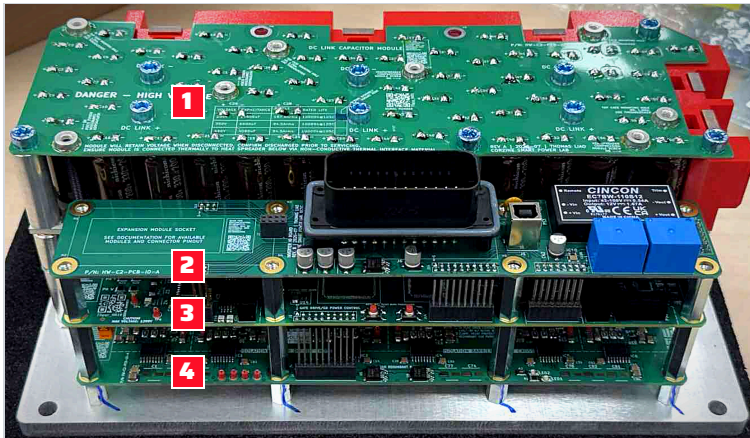


Fig. 1: The Size 2 stack, top to bottom: capacitor board, I/O board, control board, and gate-drive board.

1 DC-link capacitor board

Electrolytic cans are thermally bonded to the baseplate, with film and ceramic capacitors beneath. Multilayer ceramic capacitors (MLCCs) mount directly between the IGBT terminals. The short commutation loop limits switching overshoot and supports a future transition to SiC.

2 I/O board

USB-B, dual CAN FD, and an expansion socket for application I/O.

3 Control board, dual-MCU

FOC motor control runs on the main MCU. An independent safety coprocessor monitors every safety-critical signal and can bring the system to a safe state.

4 Gate-drive board

Isolated gate drive and sensing for the three IGBT half-bridges.

COOLING

Three IGBT half-bridge modules are conduction-cooled through an aluminum baseplate. Size 2 requires no liquid-cooling loop.

SOFTWARE AND WORKFLOW

The same control module runs every chassis. It supports motor self-commissioning for PMSM and induction machines, the open-source RTE Studio model-based workflow, and the browser-based Telemetry Viewer. RTE Studio requires no MATLAB; a software-in-the-loop pipeline is in progress. The controls program has benchmarked ADRC (active disturbance rejection control) and MPCC (model predictive current control) against the production FOC controller.

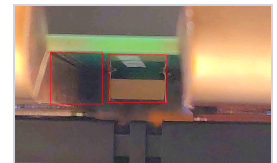


Fig. 2: Ceramic capacitors between the IGBT terminals. The commutation loop measures only a few millimeters.

PLATFORM FAMILY

One control module, four power stages

CONTROL MODULE Dual-MCU · FOC motor control + independent safety coprocessor · USB-B, dual CAN FD, and expansion I/O · RTE Studio + Telemetry Viewer

Size 1

≈50 kW class inverter for light EVs and lab benches

■ **EARLY DESIGN**

Size 2 inverter

≈100 kW · 400 V · 465 A · bench-tested

■ **IN VALIDATION**

Size 2 DC-DC

Bidirectional DC-DC for charging, storage, and microgrids

■ **IN PROGRESS**

Size 3

Megawatt class · 900 V · 1,400 A · liquid-cooled; arc-flash containment analysis in progress

■ **IN DESIGN**

EVIDENCE

Bench-tested, on the record

400 V class 150, 200, or 400 V by DC-link selection	465 A Continuous · 600 A peak (60 s)	10 Published test reports, raw telemetry linked	PMSM + induction Instrument-verified motor self-commissioning
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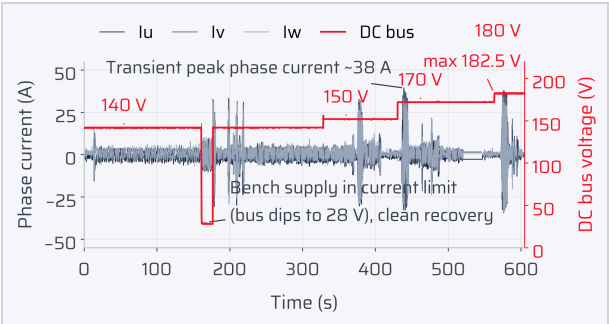


Fig. 3: 180 V power-stage bring-up. A Size 2 chassis (200 V-class DC link) drove a TECO MAX-IE3 10 HP induction motor as the bus stepped through 140, 150, 170, and 180 V. At approximately 160 s, the supply reached its current limit; after the limit was raised, the controller recovered without a reset or fault. Peak phase current was approximately 38 A; maximum bus voltage was 182.5 V.

TEST CAMPAIGN	SETUP	KEY RESULT
180 V power-stage bring-up	TECO 10 HP induction motor; bus stepped from 140 to 180 V	200 V-class operation confirmed; ≈38 A peak; clean fault recovery
One-hour no-load endurance	Size 2 + induction motor; 120 V DC; ≈60 Hz	60 minutes of continuous operation; zero faults; thermal baseline established
20-minute reversal stress at 180 V	±60 Hz every ≈45 s; passive cooling	26 reversals; zero faults; ≈41 A maximum
Self-commissioning accuracy	Inverter estimates versus LCR meter; PMSM + induction	Resistance error from -6.5 to +7.3 %; inductance error of +2.4 %

Six additional instrumented calibration reports support these results. All ten reports include raw telemetry and are revised publicly as validation continues.

SAFETY ENGINEERING

Six independent paths to the safe state

The control module is dual-MCU by design. The main MCU (STM32H723ZG) runs FOC motor control, plausibility checks, and fault handling. The safety coprocessor (STM32G474RCTx) independently monitors every safety-critical signal, cross-checks both CAN buses, and runs a challenge-response watchdog. Either processor can force the safe state.

Hardware PWM break (TIM1)	<100 ns	Shared driver RESET	<1 μs
1oo2 (one-out-of-two) gate-drive power kill	~10 μs	Independent coprocessor kill	<10 μs
Watchdog-forced main-MCU reset	~100 ms	Passive 3.3 V rail-loss pull-down	passive, no software

ASIL D design target for the highest safety goals, using ASIL B(D) decomposition: two independent ASIL B channels satisfy an ASIL D goal. ISO 26262 methodology is applied as a design input, not presented as a certification claim. The functional-safety roadmap includes fault-injection, thermal, and vibration campaigns.

17 hazards identified	>	14 safety goals	>	22 functional safety requirements	>	92 fault-injection tests planned	>	ISO 26262 + IEC 61800-5-2 safe-function mapping
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Safety engineered from the silicon to the sheet metal.

A university lab with a megawatt roadmap

OpenVVF is built in the Smart Power Lab at UC Santa Cruz. The lab's mission is to advance open power electronics by publishing hardware, firmware, and safety engineering that others can study, build, and extend.

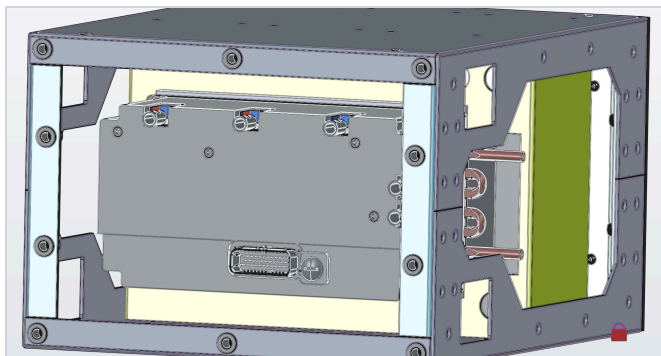
The team publishes what it learns: schematics and PCB releases, control firmware, hazard and threat analyses, and the raw telemetry behind every test claim. Sponsors and partners provide the resources for bench time, fabrication runs, and test campaigns.

SUPPORTED BY

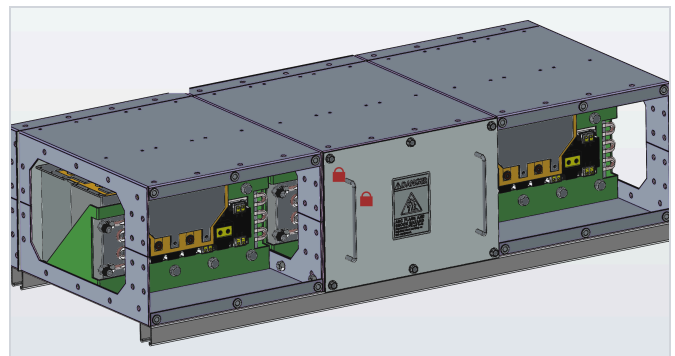


TRAJECTORY

■ IN DEVELOPMENT	■ PLANNED	■ IN PROGRESS	■ IN PROGRESS	■ IN DESIGN
Next control-module revision (Gen 7.1) Hardware revision of the dual-MCU control module.	Loaded and dynamometer testing Dynamometer testing of Gen 7.1 will take the platform from no-load validation to rated power.	Zero DSR integration A Size 2 chassis is being readied for our 2018 Zero DSR electric motorcycle.	Size 1 + Size 2 DC-DC An approximately 50 kW chassis is in early design, along with a liquid-cooled bidirectional DC-DC variant for charging, storage, and microgrids.	Size 3 megawatt-class platform The platform targets a 900 V bus and 1,400 A, with an all-film DC link and a modular liquid-cooled enclosure. Arc-flash containment analysis is in progress.



Size 2 DC-DC enclosure: Design rendering. Reuses the Size 2 power stage and control module in a liquid-cooled enclosure.



Size 3 megawatt-class enclosure: Design rendering. The enclosure features modular liquid-cooled bays; arc-flash containment analysis is in progress.

Every schematic, line of firmware, safety analysis, and test trace is public at docs.openvvf.org.

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